

Modular, Explainable, and Causal AI for Alzheimer's Risk: A Cross-Cultural Review of Lifestyle and Clinical Factors in Japanese and Indian Cohorts

¹Naina

¹B.tech Student

¹Department of Computer Science and Engineering, St. Soldier Group of Institutions, IKG
Punjab Technical University, Jalandhar, Punjab, India

Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder shaped by interacting lifestyle, behavioral, environmental, and clinical risk factors. However, most predictive models rely heavily on Western datasets such as ADNI, which limits their applicability to Asian populations—particularly Japan and India—whose dietary patterns, aging trajectories, metabolic health, air-quality exposure, and social structures differ substantially. While classical studies have focused on biomarkers or neuroimaging, emerging evidence demonstrates the importance of multi-domain, structured (Excel-format) data including demographics, vitals, metabolic profiles, cognitive assessments, diet quality, sleep patterns, and pollution exposure. This review synthesizes cross-cultural findings from Japanese and Indian cohorts, highlighting shared and population-specific risk patterns. It further evaluates the role of machine learning (ML) models applied to tabular health data, and emphasizes the need for explainable AI (XAI) tools such as SHAP and LIME for clinical trust. Additionally, causal inference frameworks—DAGs, Structured Causal Models (SCMs), and DoWhy—are reviewed as essential methods for understanding actionable risk pathways. This paper proposes a modular, domain-wise learning framework where lifestyle, clinical, cognitive, and environmental variables are modeled independently for each country. By prioritizing interpretability, cultural adaptability, and practicality, this framework aims to improve early AD prediction and ensure equitable, context-aware healthcare solutions for both Japanese and Indian populations.

Keywords: Alzheimer's Disease, Modular Machine Learning, Explainable AI, Causal Inference, Indian and Japanese Cohort

1. Introduction

1.1 What is Alzheimer's Disease?

Alzheimer's disease (AD) is a progressive and irreversible neurodegenerative disorder that impairs memory, cognition, behavior, and daily functioning. It accounts for more than 60% of global dementia cases[1]. Pathological hallmarks include extracellular amyloid- β (A β) plaque deposition, intracellular neurofibrillary tangles formed by hyperphosphorylated tau, synaptic dysfunction, and gradual neuronal loss[2]. These changes lead to atrophy in regions essential for emotional regulation, learning, and memory retrieval[3].

The burden of AD is rising rapidly in aging populations such as Japan and in fast-urbanizing countries like India. Both nations are experiencing demographic transitions, extended life expectancy, and lifestyle shifts that increase vulnerability to cognitive decline. Despite decades of biomedical investigation, current treatments offer only symptomatic relief, making early prediction, lifestyle-based risk assessment, and personalized prevention strategies essential[4].

1.2 Why Multi-Factor (Lifestyle–Clinical) Analysis is Needed

Traditional AD research has focused on amyloid- or tau-based hypotheses, but repeated therapeutic failures highlight the need for broader, multi-factor assessment. Current evidence shows that lifestyle and environmental patterns play major roles in modulating AD risk—even more significantly than genetic predisposition in many populations[5].

Key non-genetic contributors include:

- Diet patterns (high sugar, low fiber, low omega-3 intake)
- Physical inactivity
- Poor sleep quality

- Chronic stress
- Cardiovascular and metabolic health (hypertension, diabetes)
- Social isolation
- Air pollution and urban exposure[6], [7].

These factors interact biologically through inflammation, metabolic dysfunction, oxidative stress, and neuroimmune pathways. Even gut-brain interactions—previously explored primarily through microbiome studies—have shown strong modulation by diet, stress, and lifestyle, especially in Japan and India[8],[4].

Because lifestyle variables vary widely across cultures, regions, and socioeconomic groups, AD is now recognized as a multi-domain, context-dependent condition. This shift requires machine learning models capable of integrating structured clinical and lifestyle data into transparent, actionable insights[9],[10].

1.3 The Role of Cross-Cultural Data (India & Japan)

One major limitation in Alzheimer’s research is the overwhelming dependence on Western datasets such as ADNI, which do not capture the lifestyle, dietary, environmental, and social characteristics of Asian populations[10],[6]. Japan and India represent two contrasting yet highly informative contexts.

Japan

- Oldest population globally[11]
- Longitudinal aging cohorts
- Diet rich in fish, fermented foods, and fiber
- Low air pollution
- Strong social systems and community engagement
- Extensive lifestyle- and diet-focused AD studies[12]

India

- Rapid urbanization
- High genetic and cultural diversity
- Dietary transition toward processed foods
- Elevated air pollution levels and urban stress[11],[13]
- Underdeveloped AD datasets and minimal longitudinal tracking[14],[15]

Despite these complementary strengths and challenges, comparative Japan–India Alzheimer’s studies are extremely limited. Cross-cultural analysis can reveal population-specific risk patterns, culturally mediated lifestyle effects, and environmental exposures relevant to ML-based prediction models[7].

1.4 Objective of the Review

This review aims to synthesize current knowledge on lifestyle, behavioral, clinical, cognitive, and environmental factors contributing to Alzheimer’s disease risk across Japanese and Indian populations. Unlike previous reviews dominated by genomics or multi-omics perspectives, this work focuses on structured health data—demographics, vitals, metabolic indicators, diet, activity, sleep, cognition, and pollution exposure.

The review further evaluates how machine learning, especially explainable AI (SHAP, LIME) and causal inference frameworks (DAGs, SCMs, DoWhy, counterfactuals), can enhance understanding of culturally specific AD risks[10],[16],[17]. Finally, it proposes a modular, domain-wise ML framework aligned with real-world data availability in Japan and India. This approach supports transparent, ethical, and population-tailored prediction models that move beyond Western-centric trends and strengthen global equity in Alzheimer’s research.

2. Literature Review

2.1 Lifestyle, Behavioral, and Clinical Factors

While genetic markers such as APOE ε4 have long been associated with increased Alzheimer’s disease (AD) susceptibility[1], recent Asian studies demonstrate that lifestyle, behavioral, and clinical factors play equal—or greater—roles in shaping risk across culturally diverse populations. In Japan and India, AD vulnerability shows a strong correlation with modifiable conditions, including hypertension, diabetes, obesity, nutritional patterns, stress, sleep quality, and physical inactivity[6],[11].

Clinical parameters such as blood pressure, HbA1c, BMI, lipid profile, inflammatory markers (CRP, IL-6), and cognitive assessments (MMSE, MoCA) have emerged as robust early indicators of cognitive decline in aging cohorts[18],[19]. These structured measures are more accessible and culturally relevant than genetic screening in Asian contexts.

Studies across Japan highlight the protective effects of high-fiber diets, fish consumption, social engagement, and active lifestyles in reducing cognitive decline[12]. Conversely, Indian cohorts show increased prevalence of metabolic disorders, processed food intake, and chronic stress due to rapid urbanization—factors known to influence AD progression[11],[13].

Table 1. Key Lifestyle & Clinical Factors Linked to Alzheimer’s Disease

Factor	Type	Role in AD	Region Focus
Physical inactivity	Lifestyle	↑ Cognitive decline	Both
Hypertension / HbA1c	Clinical	↑ Vascular + metabolic risk	India
Stress / Cortisol	Behavioral	↑ Neuroinflammation	India
Diet quality (fiber, fish)	Lifestyle	↓ Inflammation, ↑ cognitive health	Japan

2.2 Diet Patterns, Metabolic Health, and Gut–Brain Interaction

Although microbiome studies are increasing globally, in Asia the **diet→metabolism→brain** pathway is more extensively explored than microbiome sequencing itself. Japanese research highlights that high-fiber, fermented, and omega-3–rich diets correlate with improved cognitive resilience, reduced inflammation, and better metabolic stability[12],[21]. Seasonal dietary variation in Japan has also been linked to fluctuation in cognitive symptoms among older adults[21].

Indian cohorts show contrasting trends: refined carbohydrate intake, low fiber diets, rising obesity, vitamin deficiencies, and metabolic syndrome are increasingly prevalent contributors to cognitive decline[11], [15]. Environmental and dietary transitions in urban India have been linked to elevated systemic inflammation—an established AD risk mechanism[6].

Table 2. Diet & Lifestyle Patterns Influencing AD in Asian Populations

Factor	Function	Trend in AD	Region
High-fiber, fermented foods	Supports metabolic health	Protective	Japan
Processed food intake	↑ Inflammation & oxidative stress	Increases risk	India
Chronic stress exposure	↑ Cortisol, ↓ memory	Higher	India
Regular physical activity	Enhances cognitive reserve	Protective	Both

2.3 Lifestyle & Environmental Risk Factors

Lifestyle variables—smoking, alcohol use, physical inactivity, poor sleep—and environmental exposures (air pollution, industrial toxins, noise stress) exert compounding effects on cognitive aging[6].

India: Urban regions face severe PM2.5 exposure, traffic pollution, heat stress, and industrial toxicity—all linked to neuroinflammation and accelerated cognitive decline[11], [17].

Japan: Although cleaner environments and active aging cultures help reduce risk, social isolation and low birth rates create new vulnerabilities for the elderly population[22].

Clinical markers such as CRP, cortisol, and lipid ratios show strong interactions with lifestyle and environment, demonstrating the need for multi-domain, structured-data models rather than single-factor approaches.

2.4 Clinical Biomarkers & Cognitive Scores

In Japan and India, non-invasive, structured clinical biomarkers are more feasible than CSF or stool sampling for population-level AD studies. The most consistently predictive markers include:

- Blood pressure and vascular indicators
- Fasting glucose and HbA1c
- Lipid levels (LDL, HDL, triglycerides)
- BMI and waist–hip ratio
- Inflammatory markers (CRP, IL-6)

- Cognitive scores (MMSE, MoCA)

These indicators are widely available in hospital records and excel-format datasets, making them ideal for machine learning pipelines in culturally diverse settings[18], [19].

Merging lifestyle variables with these biomarkers provides a robust foundation **for** early AD prediction in Asian cohorts, where high-cost imaging or genetic tests are less accessible.

2.5 Need for Modular Multi-Domain Modeling

Traditional integrative AD studies often rely on complex multi-omics datasets, but these are expensive, difficult to collect, and rare in India and Japan. In contrast, modular multi-domain modeling—treating lifestyle, clinical, cognitive, and environmental domains independently—provides a practical, scalable alternative[23].

Benefits include:

- Handles missing data across domains
- Respects cultural differences in lifestyle patterns
- Allows cross-cultural comparison (Japan vs India)
- Enables better interpretability

Tools such as PCA, clustering, and domain-specific ML pipelines are increasingly used to identify meaningful risk patterns in structured datasets[24].

2.6 Machine Learning for AD Prediction (Excel-Based Models)

Machine learning plays a central role in AD research using structured datasets. Logistic Regression, XGBoost, Random Forest, SVM, and LightGBM have all demonstrated strong performance on tabular clinical and lifestyle data[19].

However, most models are trained on Western datasets. Japan and India lack large-scale structured AD datasets, creating a need for country-specific pipelines integrating demographics, vitals, cognition, and lifestyle variables[13].

Table 3. ML Models for Tabular AD Prediction

Algorithm	Data Type Used	Interpretability	Applied XAI
Random Forest	Clinical + lifestyle	Moderate	SHAP
XGBoost	Clinical + cognition	High	SHAP
SVM	Imaging + vitals	Low	None
Logistic Regression	Demographics + MMSE	High	Coefficients

2.7 Explainable AI (XAI) Applications

Explainable AI tools like SHAP and LIME are crucial for interpreting ML-based AD predictions, especially in clinical deployment[25].

- **SHAP** identifies global & local feature contributions
- **LIME** explains individual prediction cases
- **Counterfactual explanations** support model auditing and patient counseling[16]
- **Surrogate models** improve interpretability for tree-based ensembles[26]

In Japan and India, where clinicians prefer transparent models, XAI bridges the gap between prediction accuracy and medical trust[27].

2.8 Causal Inference in AD Research

Causal inference moves beyond correlation to identify actionable pathways, answering Questions such as: “Would reducing pollution or improving sleep lower AD risk?”

Tools include:

- DAGs (Directed Acyclic Graphs)
- SCMs (Structural Causal Models)
- DoWhy
- CausalNex
- Temporal Causal Graphs for longitudinal data[17], [28]

Recent work highlights a lack of causal modeling in Asian AD studies, despite its importance for public health planning in Japan and India[29].

3. Cross-Cultural Comparison: Japan vs India

3.1 Japanese Cohorts: Strengths & Focus

Japan has the world’s most rapidly aging population, leading to extensive longitudinal datasets focused on cognitive decline, lifestyle behaviors, nutrition, and social engagement[12]. Japanese cohort studies benefit from strong healthcare networks, regular medical checkups, and standardized data collection, enabling reliable tracking of metabolic health, sleep quality, physical activity, and cognitive scores across older adults.

Research in Japan reveals the role of healthy aging practices—high dietary fiber intake, fish consumption, fermented foods, low red-meat diets, and community-based physical activity—in maintaining cognitive resilience[12]. Studies also highlight environmental advantages such as lower pollution exposure, stable social structures, and preventive health culture.

Although some studies examine molecular markers such as INPP5D expression changes[20], the majority of actionable insights emerge from lifestyle and clinical variables rather than genetics alone. Japanese datasets are rich in behavioral and environmental attributes but still show limited use of advanced machine learning, causal inference, or explainable AI, especially for comparison with other Asian populations.

Thus, Japan provides high-quality, long-term clinical and lifestyle datasets but lacks cross-cultural modeling and XAI-integrated prediction frameworks.

3.2 Indian Cohorts: Gaps & Potential

India provides a contrasting landscape, shaped by rapid urbanization, diverse dietary habits, socioeconomic variation, and high environmental stress. AD research in India is emerging but remains limited by small cohort sizes, inconsistent follow-up, and insufficient integration of lifestyle, clinical, and cognitive data[11], [15].

Indian elders face several risk-enhancing patterns:

- High exposure to urban pollution and PM2.5
- Increasing consumption of processed foods
- Rising rates of diabetes, hypertension, obesity
- Irregular sleep cycles and chronic stress
- Reduced physical activity in urban areas

These factors contribute significantly to neuroinflammation and cognitive decline but are underrepresented in national-level datasets.

Interestingly, Indian sociocultural features—such as multilingualism and extended family structures—show cognitive benefits and slower decline, but such protective factors are rarely studied in ML-based frameworks[32].

India’s greatest strength lies in its large, diverse population and growing digital health infrastructure. However, the lack of standardized, modular datasets limits machine learning adoption, explainability, and causal inference applications.

Developing structured lifestyle–clinical datasets and XAI-friendly predictive models will allow India to fill major global gaps.

Table 4. Summary of Differences & Research Gaps (Lifestyle–Clinical Focus)

Feature	Japan	India
Cohort Structure	Longitudinal, clinical, lifestyle-focused	Limited, cross-sectional, fragmented
Clinical Data	High quality, standardized	Inconsistent, varied by region
Lifestyle Patterns	Fiber-rich diet, high activity	Processed diet transition, low activity

Environmental Exposure	Low pollution, stable conditions	High pollution, urban stress
Research Gaps	Limited causal/XAI modeling	Limited cohort depth & data integration
Opportunities	Deep aging & lifestyle datasets	Diverse population, scope for XAI/causal frameworks

3.3 Gaps in Literature

Despite increasing AD research, major gaps remain—especially for explainable, culturally adaptable ML frameworks in Asian populations:

3.3.1 Overreliance on Black-Box Models

Many AD studies rely on deep learning or ensemble models without explainability, reducing clinical trust. SHAP and LIME are increasingly used but rarely in Asian datasets[5], [10].

3.3.2 Lack of Modular or Domain-Wise Modeling

Existing models assume complete multimodal datasets, which is unrealistic in India and Japan. Few studies treat lifestyle, clinical, cognitive, and environmental domains independently, limiting applicability across diverse cohorts[33].

3.3.3 Minimal Use of Causal Inference Tools

Causal models such as SCMs, DAGs, DoWhy, and CausalNex are underused in Asia, despite their importance in identifying actionable risk reduction pathways[34].

3.3.4 Underrepresentation of Asian Populations

AD prediction research is dominated by Western datasets (ADNI, ROSMAP). Japanese and Indian cohorts are vastly underutilized, and cultural factors like diet, pollution, multilingualism, and social engagement remain overlooked[4].

Table 5. Key Literature Gaps

Gap Area	Description
Model Transparency	Overuse of black-box AI; limited SHAP/LIME use in Asian cohorts
Data Structure	Lack of modular pipelines for lifestyle, clinical, cognitive domains
Causality	Sparse use of DAGs, SCMs, and counterfactual modeling
Cultural Representation	Western datasets dominate; minimal India/Japan integration

4. Proposed Framework

To address the limitations of Western-centric datasets and improve Alzheimer’s prediction in Asian populations, we propose a modular, interpretable machine learning framework tailored for Japanese and Indian cohorts. Prior studies highlight that lifestyle, metabolic, environmental, and cognitive factors strongly shape AD risk across Asian populations [15], [32], making structured data a practical alternative to expensive multi-omics or imaging approaches [1], [2], [35].

4.1 Domain-Wise Data Modules

The framework separates variables into four modules to handle heterogeneous data availability:

- **Lifestyle module:** diet, physical activity, sleep, and stress. Japanese studies show protective effects of high-fiber and fermented diets [12], [21] while Indian data link processed food, pollution, and stress to cognitive decline [11], [13]
- **Clinical module:** BP, HbA1c, BMI, lipid profile, inflammation markers—highly predictive and widely available in tabular datasets [36],[18], [19].
- **Cognitive module:** MMSE, MoCA, memory tests, which vary cross-culturally [32].
- **Environmental module:** pollution and heat exposure, critical in India [17]

4.2 Modular ML Pipelines

Each module is processed with independent ML pipelines (e.g., Logistic Regression, Random Forest, LightGBM, hybrid models [19],[37] consistent with modular approaches recommended by AD structured dataset1718.This enables flexible modelling, better handling of missing data, and adaptation to country-specific variables.

4.3 Explainable AI Layer

SHAP and LIME provide transparent feature-level and case-level explanations [38] while surrogate models and counterfactuals support clinician interpretability [16], [26], [27]. This increases trust and usability in real clinical settings.

4.4 Causal Inference Integration

Causal tools, such as DAGs, SCMs, and DoWhy, help identify modifiable risk pathways and disentangle confounders. [10], [36], [28], [29] This supports actionable insights like the causal impact of diet, sleep, or pollution on AD risk.

4.5 Cross-Cultural Comparison

Parallel modeling for Japanese and Indian cohorts enables detection of shared vs. population-specific risk patterns, addressing the underrepresentation of Asian datasets in AD research [24], [34].

4.6 Lightweight Dashboard

A simple Streamlit dashboard visualizes predictions, SHAP values, and cultural comparisons, enabling practical use in preventive health programs.

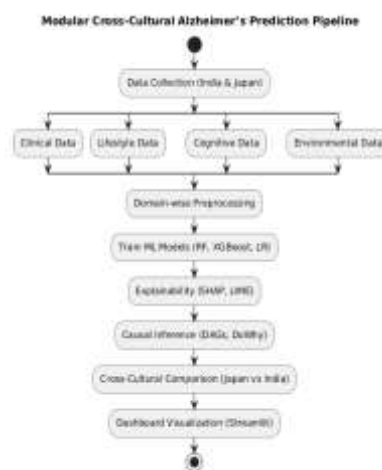


Fig 1. Model pipeline flowchart

5. Conclusion

This review shows that culturally informed and interpretable machine learning frameworks are urgently needed for Alzheimer's prediction in Japan and India. Existing research relies heavily on western datasets and single-domain analysis, leaving major gaps in understanding how lifestyle, clinical, and environmental factors shape risk in Asian populations. To address this, we proposed a modular, domain-wise learning approach where each data type—clinical markers, lifestyle habits, cognition, and environmental exposures—is modeled independently for each country. Integrating SHAP, LIME, and causal inference tools such as DAGs and SCMs ensures that predictions are both transparent and clinically meaningful. The resulting cross-cultural dashboard enables comparison of risk factors between Japanese and Indian cohorts and supports tailored, population-specific interventions. This framework is practical, scalable, and aligned with ethical precision-medicine goals, offering an immediately actionable path toward more inclusive and culturally relevant AD prediction models.

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